

Module 5 Unit testing

HandsOn 2: Django Models

Migrations:

```
C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module5_PythonBackendFrameworks\HandsOn_2\trainingmanager>python manage.py makemigrations trainer
No changes detected in app 'trainer'

C:\Users\vigne\Desktop\Cognizant_Deep_Skilling\Module5_PythonBackendFrameworks\HandsOn_2\trainingmanager>python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, sessions, trainer
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying sessions.0001_initial... OK
  Applying trainer.0001_initial... OK
```

Insertions:

```
In [7]: cs = Department.objects.create(dept_name="Computer Science", hod_name="Dr. Alan Turing", budget=500000.00)
In [8]: ee = Department.objects.create(dept_name="Electrical Engineering", hod_name="Dr. Nikola Tesla", budget=500000.0
Out[8]: <Department: Electrical Engineering>
In [9]: m1 = SkillModule.objects.create(module_name="Python Backend", module_code="CS101", credits=4, department=cs)
In [10]: m2 = SkillModule.objects.create(module_name="Data Structures", module_code="CS102", credits=4, department=cs)
In [11]: m3 = SkillModule.objects.create(module_name="Circuit Analysis", module_code="EE101", credits=3, department=ee)
Out[11]: <SkillModule: Circuit Analysis>
In [12]: m4 = SkillModule.objects.create(module_name="Digital Electronics", module_code="EE102", credits=3, department=
Out[12]: <SkillModule: Digital Electronics>
In [13]: Employee.objects.create(first_name="Alice", last_name="Smith", email="alice@corp.com", department=cs, hiring_y
Out[13]: <Employee: Alice Smith>
In [14]: Employee.objects.create(first_name="Bob", last_name="Jones", email="bob@corp.com", department=cs, hiring_year=
Out[14]: <Employee: Bob Jones>
In [15]: Employee.objects.create(first_name="Charlie", last_name="Brown", email="charlie@corp.com", department=ee, hirin
Out[15]: <Employee: Charlie Brown>
In [16]: Employee.objects.create(first_name="Diana", last_name="Prince", email="diana@corp.com", department=ee, hiring_
Out[16]: <Employee: Diana Prince>
In [17]: Employee.objects.create(first_name="Evan", last_name="Wright", email="evan@corp.com", department=cs, hiring_ye
Out[17]: <Employee: Evan Wright>
In [18]: cs_modules = SkillModule.objects.filter(department__dept_name='Computer Science')
In [19]: print(cs_modules)
Out[19]: <QuerySet [(<SkillModule: CS101 - Python Backend>, <SkillModule: CS102 - Data Structures>)]>
In [20]: connection.queries_log.clear()
In [21]: employees=list(Employee.objects.select_related('department'))
In [22]: print(connection.queries[-1])
Out[22]: {'sql': 'SELECT "trainer_employee"."id", "trainer_employee"."first_name", "trainer_employee"."last_name", "trainer_employee"."email", "trainer_employee"."date_of_birth", "trainer_employee"."hiring_year", "trainer_employee"."is_active", "trainer_employee"."department_id", "trainer_department"."id", "trainer_department"."dept_name", "trainer_department"."hod_name", "trainer_department"."budget" FROM "trainer_employee" LEFT OUTER JOIN "trainer_department" ON ("trainer_employee"."department_id" = "trainer_department"."id")', 'time': '0.000'}
```

Unique together constraint:

Add certification

Please correct the error below.

Certification with this Employee and Skill module already exists.

Completion date:

2026-07-08

Today 

Note: You are 5.5 hours ahead of server time.

Grade:

A

Employee:

Evan Wright

 

Skill module:

PY001 - Advanced python

 